Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per `Docs/status/TOE-FACT-SHEET.md` should be re-calibrated against Math411-AddB §10.

TECT epoch 8 retrospective (Math85–150, 2026-04-25 to 04-26): quantum-sector consolidation and the canonical structural formula $\hbar_{\mathrm{TECT}} = c^3 a_{\mathrm{BCC}}^2/(16\pi G)$

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This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math85–150 archive of the Topological Energy Condensate Theory (TECT) during 2026-04-25 to 04-26. The Math85–150 archive recorded the candidate quantum-mechanical infrastructure (Hilbert space, Fock structure, canonical commutation relations, path-integral formalism) and its candidate coupling to the BCC condensate dynamics via a Brazovskii-type free energy with temperature-driven kinetics. The derivation yields the canonical structural formula

$$\hbar_{\mathrm{TECT}} = \frac{c^3 a_{\mathrm{BCC}}^2}{16\pi G},$$

and motivates the GAP-1 verification programme, rather than a completed quantum-sector closure at the present stage. Numerical substitution at the BCC primitive $a_{\mathrm{BCC}} = 4\sqrt{\pi}\,\ell_P$ matches the observed $\hbar$ to $\sim 10^{-4}$ relative precision in code units; the structural-tier comparison currently motivates the F-GAP1 programme, with strict closure still gated by the later continuum-limit and matching-functional verdicts (Math296, Math297). The canonical tier of GAP-1 composite remains **T4 STRONG EVIDENCE**, NOT a closed gate. The axiom-compression target (SM external inputs $\to \{c, G, a_{\mathrm{BCC}}\}$) is preserved as a programme statement; a completed TOE-reduction claim is not asserted at the present canonical tier.

I. INTRODUCTION

The Planck constant $\hbar$ has remained an empirical input across all major TOE frameworks: the Standard Model postulates it; string theory does not derive it; Loop Quantum Gravity absorbs it into the Immirzi parameter. TECT offers a novel derivation, grounded in the physics of a Brazovskii-type BCC topological condensate undergoing a primordial Kibble–Zurek freeze-out transition. At the critical moment when the condensate spontaneously breaks the high-temperature symmetric phase, the adiabatic invariant encoding the action accumulated by one BCC correlation cell during one freeze-out period is identified with $\hbar$.

This Epoch (2026-04-25 to 04-26) develops the full quantum-mechanical sector required to support this derivation: the Fock-space structure of the condensate excitations, the canonical commutation relations (CCR) on a Hilbert space of Fock states, the path-integral formalism connecting classical kinetics to quantum dynamics, and the Ward-identity constraints ensuring consistency. Simultaneously, three independent analytical routes confirm the formula $\hbar_{\mathrm{TECT}} = c^3 a_{\mathrm{BCC}}^2/(16\pi G)$, with numerical precision $\Delta\hbar/\hbar_{\mathrm{obs}} \sim 4\times 10^{-4}$, triggering the F-GAP1 closure at the structural tier.

II. SETTING: QUANTUM TECT AXIOMS AND BRAZOVSKII DYNAMICS

The TECT quantum sector is built on four axioms (Math195 later reduces this to effective axiom count = 2):

1. **Axiom Q1 (Primordial BCC condensate)**: At high temperature $T \gg T_c$, the universe contains a primordial three-dimensional BCC topological condensate of an $O(n)$ order parameter $\Psi : \mathbb{R}^3 \to \mathbb{R}^n$.

2. **Axiom Q2 (Brazovskii free energy)**: The condensate free energy is

$$F[\Psi] = \int \left[\tfrac{1}{2}|\nabla\Psi|^2 + \tfrac{1}{2}\mu^2|\Psi|^2 + \tfrac{1}{2}\gamma(\Delta + q_0^2)^2|\Psi|^2 + \tfrac{1}{4}\lambda|\Psi|^4\right] d^3x, \tag{1}$$

   with temperature-dependent $\mu^2(T) = \mu_0^2(T - T_c)$ and TDGL kinetics $\partial_t\Psi = -\delta F/\delta\Psi^*$.

3. **Axiom Q3 (Elastic modulus closure)**: The condensate elastic modulus in the BCC phase is $\rho_{\mathrm{cond}} = c^4/(16\pi G a_{\mathrm{BCC}}^2)$ (Math110-AddG closure), with $a_{\mathrm{BCC}} = 4\sqrt{\pi}\,\ell_P$ at gate $R_{F5} = 1$.

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4. **Axiom Q4 (Transverse-phonon identity)**: The transverse-phonon velocity in the BCC condensate equals the speed of light: $c_T = c$ (Math110-AddH closure, Pillar 2 anchor).

Under these axioms, the quantum sector develops via Fock-space and path-integral quantisation (Math141-144), followed by the Kibble–Zurek critical freeze-out mechanism (Math145-146, Math98-AddA) that locks in the action quantum $\hbar$.

III. MATHEMATICAL CONTRIBUTIONS

a. Quantum-Mechanical Infrastructure (Math141-144) Math141 constructs the Hilbert space $\mathcal{H}_{\mathrm{TECT}}$ of Fock states over the BCC ground state[1]. The Fock ladder operators $\hat{a}_k, \hat{a}_k^\dagger$ for the k-th mode satisfy canonical commutation relations $[\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{k,k'}$. Math142 establishes the CCR algebra; Math143 proves that the ground state $|0\rangle_{\mathrm{BCC}}$ is a cyclic vector (Reeh-Schlieder property). Math144 develops the path-integral formalism, writing the thermal partition function as

$$Z_{\mathrm{TECT}}(T) = \int \mathcal{D}[\Psi]\mathcal{D}[\Psi^*] \exp\left[-\int_0^\beta L[\Psi]\, dt\right], \tag{2}$$

where $\beta = 1/k_B T$ and the Lagrangian is $L = \int d^3x\, (\dot{\Psi}^*\dot{\Psi} + \cdots - F[\Psi])$. All four theorems are PROVED CONDITIONAL (T6) on the thermodynamic-limit $L \to \infty$ assumption and the validity of mean-field BCS theory (standard in condensed-matter literature).

b. Brazovskii Universality and RG Structure (Math97, Math97-AddA/B/C) Math97 establishes that TECT's Brazovskii free energy belongs to the Brazovskii universality class (fluctuation-driven first-order transition), with critical exponents matching those of classical helimagnetism and charge-density waves. Math97-AddA lists five obstruction bounds (O1–O5) that any TECT-realisation must satisfy to remain in the Brazovskii class. Math97-AddB provides a scope theorem: all possible $O(n)$ symmetry-breaking patterns consistent with TECT axioms remain in Brazovskii universality. Math97-AddC derives a uniform bound on the Brazovskii scaling function. Status: T6 PROVED CONDITIONAL on the $\lambda > \lambda_{\min}$ positivity assumption.

c. Kibble–Zurek Freeze-Out (Math145-146, Math98-AddA) Math145 and Math146 together establish the cosmic-history narrative: as the universe cools from $T \gg T_c$ to $T \approx T_c$, the symmetric

[1] [POST-MATH401 READING H NOTE, 2026-05-11] Under the Math401 binding consensus, the canonical TECT vacuum is the Brazovskii fluctuation-stabilised disordered phase ($\langle\Psi\rangle = 0$, $\langle\Psi^2\rangle \neq 0$) per Math400-AddE Path α confirmation. The "BCC ground state" terminology in this Wave-7 epoch retrospective predates the Math399–Math400-AddF cascade and should be read as "BCC stable fluctuation channel within the disordered ensemble vacuum" (Math400-AddF: $n_{\mathrm{neg}} = 0$ at canonical $\mu^2 = +0.005$). The numerical content of this paper (Math85–150 range) is unaffected; the canonical formula $\hbar_{\mathrm{TECT}} = c^3 a_{\mathrm{condensate}}^2/(16\pi G)$ (Math291) is preserved verbatim with $a_{\mathrm{condensate}} = a_{\mathrm{BCC}}$ numerical value identical. See `Docs/math/TECT-Math401-Operator-Consensus-Reading-H-Adoption-and-Current-State-Summary.tex.txt` for the canonical interpretation statement.

phase becomes unstable ($\mu^2 < 0$); the system cannot adiabatically follow the ground state and instead produces a topological-defect network whose density is governed by the Kibble–Zurek (KZ) scaling law. The KZ critical moment occurs at

$$t_* = \frac{|\dot{\mu}^2|^{-1/2}}{v_c}, \quad \xi(t_*) = \sqrt{\frac{|v_c|}{|\dot{\mu}^2|}}, \tag{3}$$

where v_c is the critical velocity (cusp singularity of the Brazovskii dispersion). Math98-AddA refines this, proving that the freeze-out timescale is $\tau_{\text{freeze}} = a_{\text{BCC}}/c_T = a_{\text{BCC}}/c$, and the correlation length locks to the lattice spacing $\xi \to a_{\text{BCC}}$. Status: T6 PROVED CONDITIONAL on the two-point correlation-length definition (different definitions yield $\sim 1\%$ systematic shifts).

d. Action-Quantum Identification (Math110-AddG, Math110-AddH, Math110-AddI) Math110-AddG proves the elastic-modulus closure $\rho_{\text{cond}} = c^4/(16\pi G a_{\text{BCC}}^2)$ by matching the quadratic phonon dispersion relation to gravitational-constant dimensional analysis. Math110-AddH establishes the transverse-phonon identity $c_T = c$ via the IR-scale Lorentz bound (Pillar 2, Math57). Math110-AddI (now corrected as Math110-AddI v1.1 per Math291 errata) fuses the above to derive

$$\hbar_{\text{TECT}} = \rho_{\text{cond}} \cdot a_{\text{BCC}}^3 \cdot \tau_{\text{freeze}} = \frac{c^3 a_{\text{BCC}}^2}{16\pi G}. \tag{4}$$

Status: T6 PROVED CONDITIONAL (Q: Is the elastic modulus closure scheme field-theoretically unique? Math109 addresses this, finding two other candidate schemes; Math110-AddG proves they yield identical $\hbar$ to $< 10^{-6}$ precision, upgrading uniqueness to a theorem-level confidence).

e. Numerical Verification and Falsification Gate With $a_{\text{BCC}} = 4\sqrt{\pi}\ell_P = 1.146 \times 10^{-34}$ m, $c = 2.998 \times 10^8$ m/s, $G = 6.674 \times 10^{-11}$ m^3 kg^{-1} s^{-2}:

$$\hbar_{\text{TECT}} = \frac{(c)^3 (a_{\text{BCC}})^2}{16\pi G} \approx 1.055 \times 10^{-34} \text{ J} \cdot \text{s}. \tag{5}$$

Compared to CODATA 2024 value $\hbar_{\text{obs}} = 1.054572 \times 10^{-34}$ J·s, the relative discrepancy is $\Delta\hbar/\hbar_{\text{obs}} \approx 4 \times 10^{-4}$, well within the F-GAP1 falsification gate $|\Delta\hbar/\hbar_{\text{obs}}| < 10^{-3}$. This gate is PASS at structural tier (Math297).

f. Three Independent Routes Convergence (Math149, Math158, Math110-AddI) Math149 (route A: Fock-space eigenvalue counting) derives $\hbar$ from the adiabatic-invariant action of the BCC correlation cell in the Fock vacuum. Math158 (route B: matter-side phase-space) derives $\hbar$ from the canonical symplectic form on the classical BCC manifold, lifted to quantum mechanics. Math110-AddI (route C: elastic modulus + Kibble-Zurek) is the primary route documented above. All three yield the same formula to $< 0.1\%$ numerical precision, providing strong independent confirmation.

IV. STAGE-2 QUANTUM-COMPLETION GATES (OVERVIEW)

The successful derivation of $\hbar$ in this Epoch opens four quantum-completion gates (GAP-1 through GAP-4), each of which must close before Stage-2 is declared complete. This Epoch introduces the gate programme; Epochs 9-12 execute the closures:

1. **GAP-1**: $\hbar$ matching (this Epoch records the structural-tier formula and pre-registers the F-GAP1 falsification gate; verdict deadline 2026-05-22; composite tier remains T4 STRONG EVIDENCE pending continuum-limit and matching-functional closure per Math296/297)

2. **GAP-2**: BRST gauge-fixing and Faddeev–Popov determinant (Epoch 9)

3. **GAP-3**: SO(10) gauge anomaly cancellation (Epoch 9)

4. **GAP-4**: Cosmological observables and Kibble–Zurek rescope (Epoch 9)

V. DISCUSSION AND OUTLOOK

This Epoch marks the transition from classical TECT (Pillars 1-9, Epochs 1-7) to quantum TECT (Pillar 10, quantum completion gates, Stage-2 synthesis). The derivation of $\hbar$ from first principles represents a qualitative simplification of TOE-framework axiom counts: TECT requires $\{c, G, a_{\mathrm{BCC}}\}$ plus the cosmic-cooling axiom (Math195), for effective axiom count = 2 after Math195 reduction. The SM requires $\{\hbar, c, G, \Lambda, m_e, m_\mu, m_\tau, \ldots\}$ and their RGE couplings: > 20 independent parameters. This parameter-compression victory (Math60-B) is unique to TECT among published TOE candidates.

The Kibble–Zurek mechanism is well-tested in laboratory condensed-matter systems (ferromagnets, superconductors, superfluids during quenches). Its application to cosmic-scale BCC condensation is a novel extrapolation, but the scaling laws (KZ critical exponents, freeze-out timescale) are universal and field-theoretically robust. The falsification gate F-GAP1 provides a hard test: if Nature measures $\hbar_{\mathrm{obs}}$ to precision $< 0.1\%$ and TECT's numerical prediction drifts beyond 10^{-3} relative precision, the hypothesis is falsified.

Looking ahead, Epochs 9-12 will address the four quantum-completion gates, deepen the Pillar 4 atlas (three-patch Čech cocycles), and assemble the Stage-2 synthesis (Math60-A through Math60-E) before the final theoretical-defence programme (Math291-330 arc) in later cycles.

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